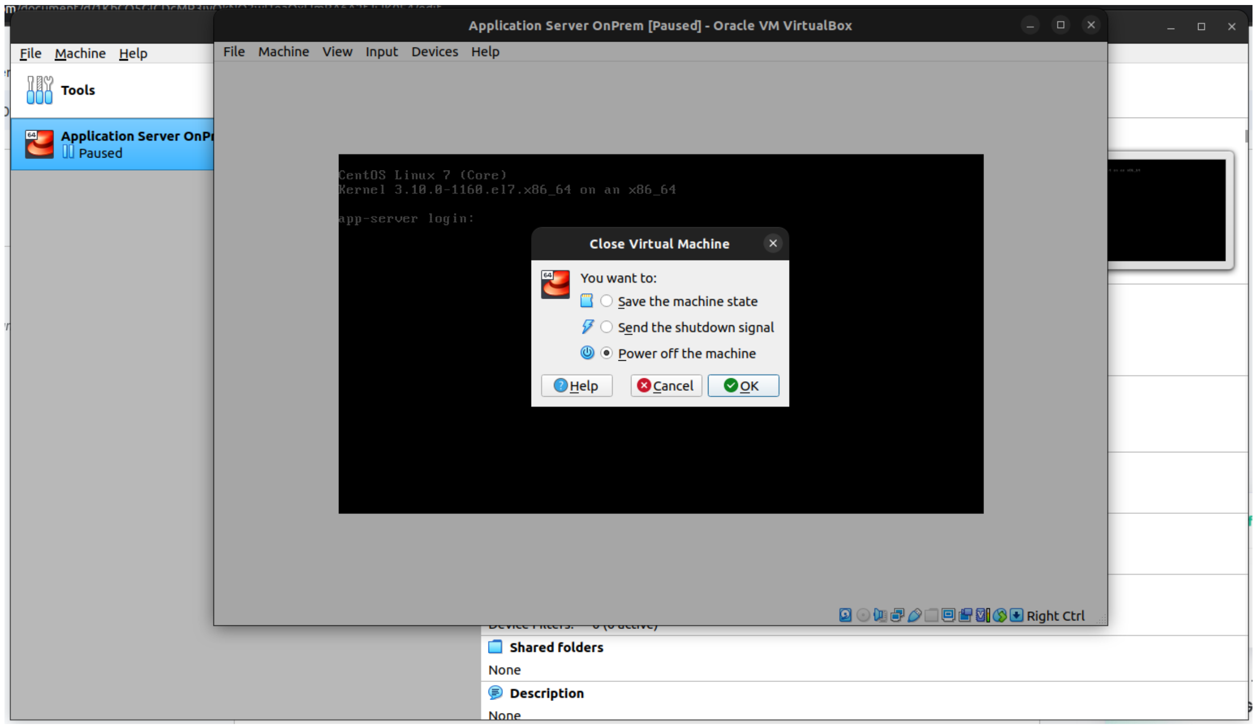
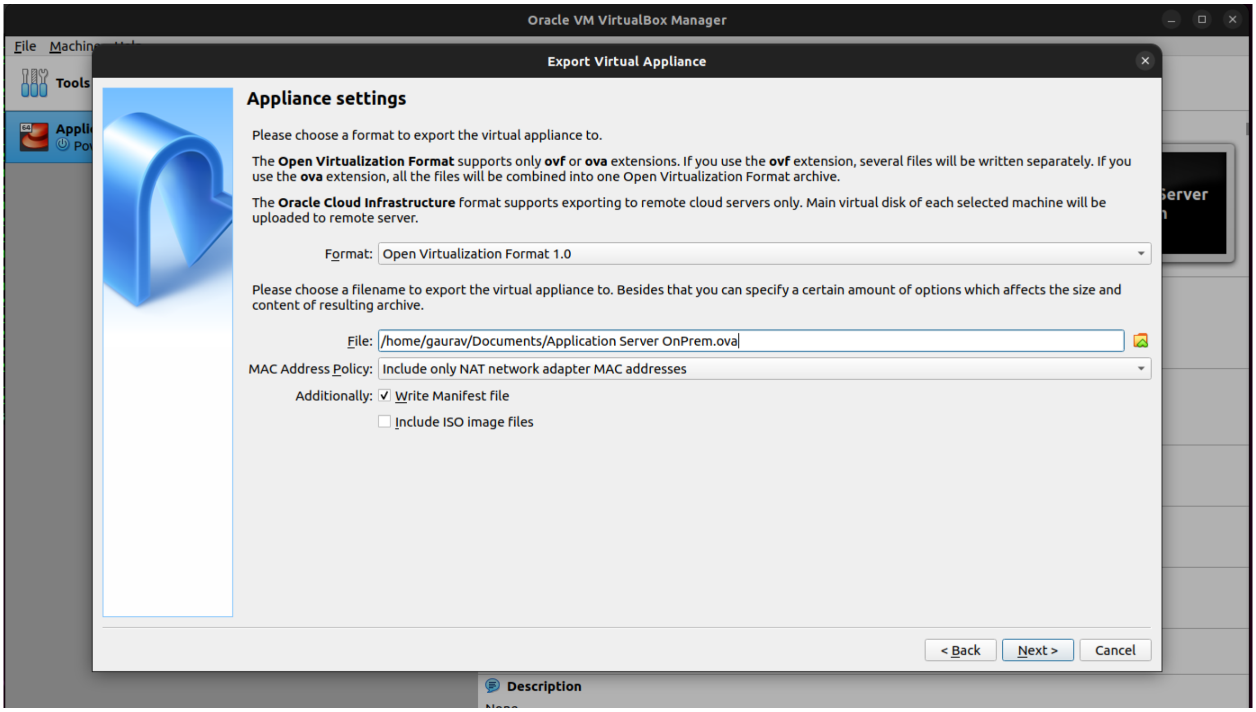
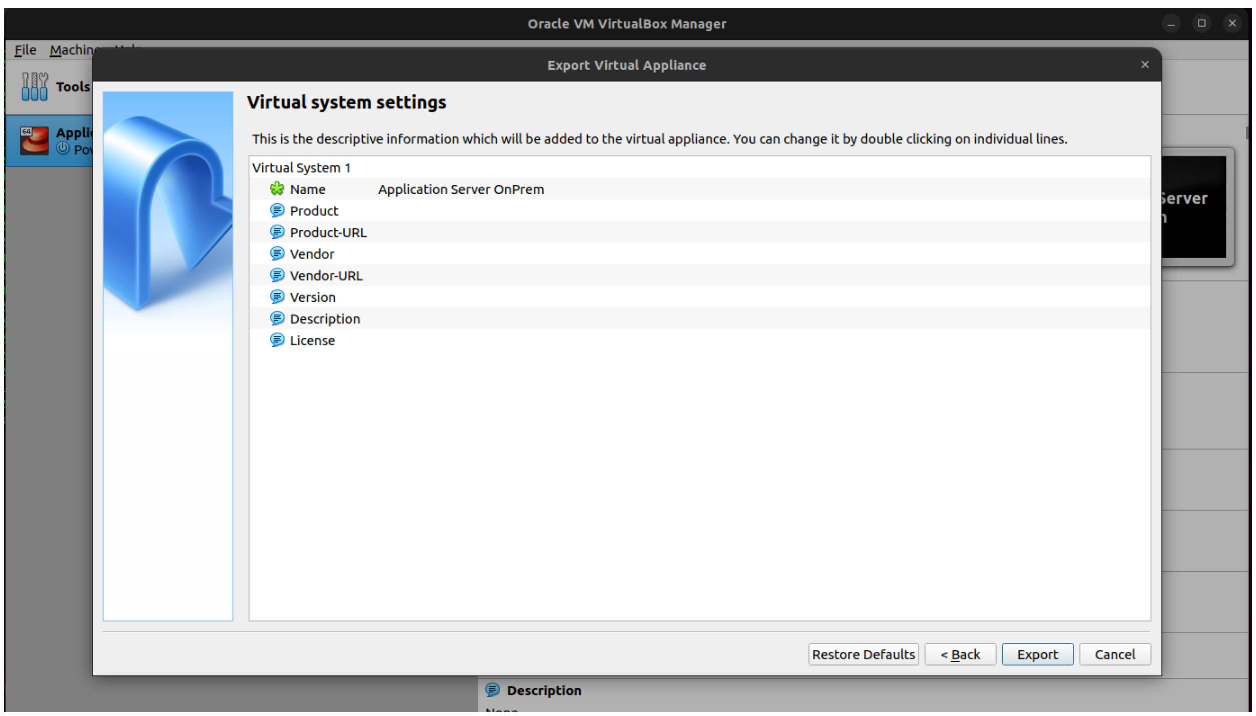
Step - 1

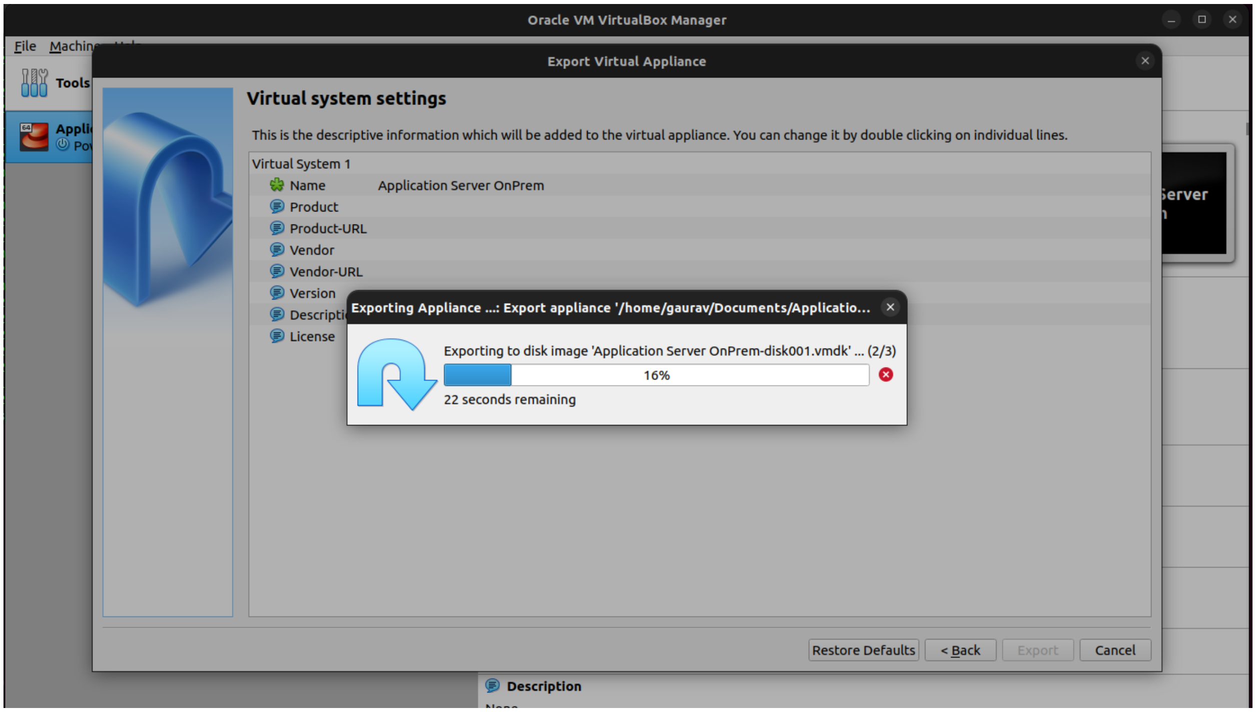
Poweroff the VM if running.



Step - 2

Right click on the VM and choose “export to OCI”


Oracle VM VirtualBox Manager
Export Virtual Appliance
Appliance settings
Please choose a format to export the virtual appliance to.
The Open Virtualization Format supports only ovf or ova extensions. If you use the ovf extension, several files will be written separately. If you use the ova extension, all the files will be combined into one Open Virtualization Format archive.
The Oracle Cloud Infrastructure format supports exporting to remote cloud servers only. Main virtual disk of each selected machine will be uploaded to remote server.
Format: Open Virtualization Format 1.0
Please choose a filename to export the virtual appliance to. Besides that you can specify a certain amount of options which affects the size and content of resulting archive.
File: /home/gaurav/Documents/Application Server OnPrem.ova
MAC Address Policy: Include only NAT network adapter MAC addresses
Additionally: Write Manifest file
Include ISO image files
< Back
Next >
Cancel
Description

Oracle VM VirtualBox Manager
Export Virtual Appliance
Virtual system settings
This is the descriptive information which will be added to the virtual appliance. You can change it by double clicking on individual lines.
Virtual System 1
Name Application Server OnPrem
Product
Product-URL
Vendor
Vendor-URL
Version
Description
License
Restore Defaults
< Back
Export
Cancel
Description



Step -3

Create a S3 Bucket in the aws region where you want the vm to be imported and upload the exported ova file to the S3 Bucket.

Step - 4

Install AWS CLI locally or Launch an AWS EC2 Instance of Amazon Linux which comes with aws cli pre-installed.

Step - 5

Create an IAM User with EC2, S3 and IAM Full access and generate IAM API keys (access key and secret access key).

Step - 6

Configure the AWS CLI with the IAM User API key created in step 5.

aws configure

Step - 7

Create an IAM Role using AWS CLI.

touch trust-policy.json

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": { "Service": "vmie.amazonaws.com" },
      "Action": "sts:AssumeRole",
      "Condition": {
        "StringEquals":{
          "sts:Externalid": "vmimport"
        }
      }
    }
  ]
}
```

```
aws iam create-role --role-name vmimport --assume-role-policy-document
"file:///home/ec2-user/trust-policy.json"
```

The above command will create an IAM role with the name vmimport.

- Create an IAM Policy for the role which will grant access to the S3 Bucket where the ova file is uploaded.

```
aws iam put-role-policy --role-name vmimport --policy-name vmimport --policy-document
"file:///home/ec2-user/role-policy.json"
```

```
touch role-policy.json
```

```
{

   "Version": "2012-10-17",

   "Statement": [

      {

         "Effect": "Allow",

         "Action": [

            "s3:GetBucketLocation",

            "s3:GetObject",

            "s3:ListBucket",

            "s3:PutObject",
```

```
            "s3:GetBucketAcl"
         ],
         "Resource": [
            "arn:aws:s3:::vmimport-123",
            "arn:aws:s3:::vmimport-123/*"
         ]
      },
      {
         "Effect": "Allow",
         "Action": [
            "ec2:ModifySnapshotAttribute",
            "ec2:CopySnapshot",
            "ec2:RegisterImage",
            "ec2:Describe*"
         ],
         "Resource": "*"
      }
   ]
}
```

Step - 7

Create a file bucket.json and update it with your bucket name and object name.

```
[
  {
    "Description": "My Server OVA",
    "Format": "ova",
    "UserBucket": {
```

```
      "S3Bucket": "vmimport-123",
      "S3Key": "Application Server OnPrem.ova"
   }
 }
]
```

- aws ec2 import-image --description "*My OnPrem VM*" --disk-containers "file:*//home/ec2-user/bucket.json*"

  ```
  {
     "Description": "My server VM",
     "ImportTaskId": "import-ami-052c4970ce867b739",
     "Progress": "1",
     "SnapshotDetails": [
         {
            "Description": "My Server OVA",
            "DiskImageSize": 0.0,
            "Format": "OVA",
            "UserBucket": {
               "S3Bucket": "vmimport-123",
               "S3Key": "Application Server OnPrem.ova"
            }
         }
     ],
     "Status": "active",
     "StatusMessage": "pending"
  }
  ```

To check import status:

- aws ec2 describe-import-image-tasks --import-task-ids import-ami-052c4970ce867b739

{

"ImportImageTasks": [

{

"Architecture": "x86_64",

"Description": "My server VM",

"ImportTaskId": "import-ami-052c4970ce867b739",

"LicenseType": "BYOL",

"Platform": "Linux",

"Progress": "39",

"SnapshotDetails": [

{

"DeviceName": "/dev/sda1",

"DiskImageSize": 797688320.0,

"Format": "VMDK",

"Status": "completed",

"UserBucket": {

"S3Bucket": "vmimport-507801817713",

"S3Key": "Application Server OnPrem.ova"

}

}

],

"Status": "active",

"StatusMessage": "booting",

"Tags": [],

```
            "BootMode": "legacy_bios"
        }
    ]
}
```

Home Work:

1) Create a VM in vmware/virtual/hyperV of ubuntu
2) Deploy a web application and mysql database
3) Migrate the vm to aws